

RFR Regime

(2. SOFR Futures and Term Rates)

CME 1M and 3M Future Quotes

ONE-MONTH SOFR FUTURES - QUOTES

VENUE: GLOBEX

  AUTO-REFRESH IS OFF Last Updated 14 Mar 2025 07:37:19 PM CT. Market data is delayed by at least 10 minutes.

MONTH	OPTIONS	CHART	LAST	CHANGE	PRIOR SETTLE	OPEN	HIGH	LOW	VOLUME	UPDATED
 MAR 2025 SR1H5			95.6775	+0.005 (+0.01%)	-	95.675	95.6775	95.675	9,874	16:37:32 CT 14 Mar 2025
 APR 2025 SR1J5			95.69	UNCH (UNCH)	-					
 MAY 2025 SR1K5			95.75	-0.015 (-0.02%)	-					
 JUN 2025 SR1M5			95.82	-0.025 (-0.03%)	-					

THREE-MONTH SOFR FUTURES - QUOTES

VENUE: GLOBEX

  AUTO-REFRESH IS OFF Last Updated 14 Mar 2025 07:44:29 PM CT. Market data is delayed by at least 10 minutes.

MONTH	OPTIONS	CHART	LAST	CHANGE	PRIOR SETTLE	OPEN	HIGH	LOW	VOLUME	UPDATED
 DEC 2024 SR3Z4			95.6325	-0.0025 (UNCH)	-	95.6325	95.635	95.6325	7,778	16:38:00 CT 14 Mar 2025
 JAN 2025 SR3F5			95.6525	UNCH (UNCH)	-	95.6525	95.6525	95.6525	17	16:38:21 CT 14 Mar 2025
 FEB 2025 SR3G5			-	-	-	-	-	-	0	16:38:22 CT 14 Mar 2025
 MAR 2025 SR3H5			95.70	-0.0075 (-0.01%)	-	95.71	95.7125	95.6975	452,084	16:38:00 CT 14 Mar 2025
 APR 2025 SR3J5			95.775	-0.005 (-0.01%)	-	95.775	95.775	95.775	810	16:38:22 CT 14 Mar 2025

<https://www.cmegroup.com/markets/interest-rates/stirs/one-month-sofr.quotes.html>

<https://www.cmegroup.com/markets/interest-rates/stirs/three-month-sofr.quotes.html>

CME Future Product Nomenclature

► Let's understand the nomenclature for product codes

JAN	FEB	MAR	APR	MAY	JUN
F	G	H	J	K	M
JUL	AUG	SEP	OCT	NOV	DEC
N	Q	U	V	X	Z

2023

Contract month.

Product Cd	Expiry	Ticker	Description
SR3	H3	SR3H3	RFR USD 3M SOFR Futures MAR-2023
SR3	M3	SR3M3	RFR USD 3M SOFR Futures JUN-2023
SR3	U3	SR3U3	RFR USD 3M SOFR Futures SEP-2023
SR3	Z3	SR3Z3	RFR USD 3M SOFR Futures DEC-2023

CME Future Basics



- CME futures are extremely liquid and are used massively for trading and hedging and building the short end ($T < 1$ year) of the yield-curve. Below are some key numbers on CME future contracts.

Ticker	Description	CCY	Contract Size	Contract Period(M)	Tick Value ¹	Initial Margin	Term Rate
ED	USD 3M LIBOR Futures	USD	1,000,000	3	25.00	2,500.00	Simple
SR1	USD 1M SOFR Futures	USD	5,000,000	1	41.67	2,500.00	Arithmetic
SR3	USD 3M SOFR Futures	USD	1,000,000	3	25.00	2,500.00	Geometric
L	GBP 3M LIBOR Futures	GBP	1,000,000	3	12.50	2,500.00	Simple
SO3	GBP 3M SONIA Futures	GBP	1,000,000	3	25.00	2,500.00	Geometric
ER	EUR 3M LIBOR Futures	EUR	1,000,000	3	25.00	2,500.00	Simple
EON	EUR 1M ESTR Futures	EUR	1,000,000	1	25.00	2,500.00	Arithmetic
IT	JPY 3M TIBOR Futures	JPY	100,000,000	3	2,500	250,000	Simple

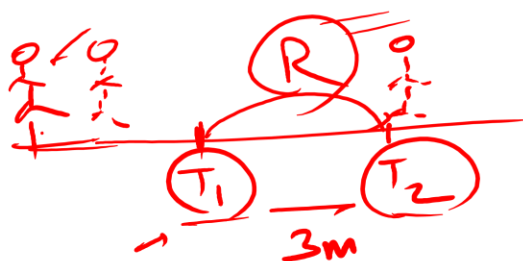
¹ The tick value per basis point (bps), also known as the basis point value (BPV)

- The ones highlighted in yellow are LIBOR based futures which are discontinued. In our examples, we will focus on SOFR 3M and SOFR 1M Futures.



Price Basis and Tick Value

- **Price basis** of the SOFR Future contract and the referenced rate are inversely proportional. The Price basis or **IMM index** given as



$$\text{Price Basis} = 100(1 - R)$$

$$\Delta P = -100(\Delta R)$$

- The **Tick value** represents the change in value of the future contract for 1 basis point change in the reference rate. It's defined as

Notional

$$\text{Tick Value} = \text{Contract Size} \times \text{Contract Period} \times 0.01\%$$

$$3/12 \times =$$

Exercise – Show that the tick value for SR3 and SR1 are \$25 and \$41.67, respectively.

Daily Variation Margin

- The Future daily variation margin calculation can be performed easily based on the daily basis point movement and the tick value applicable to the contract. Below is the PnL for a SR3 future contract

Future Daily Variation Margin

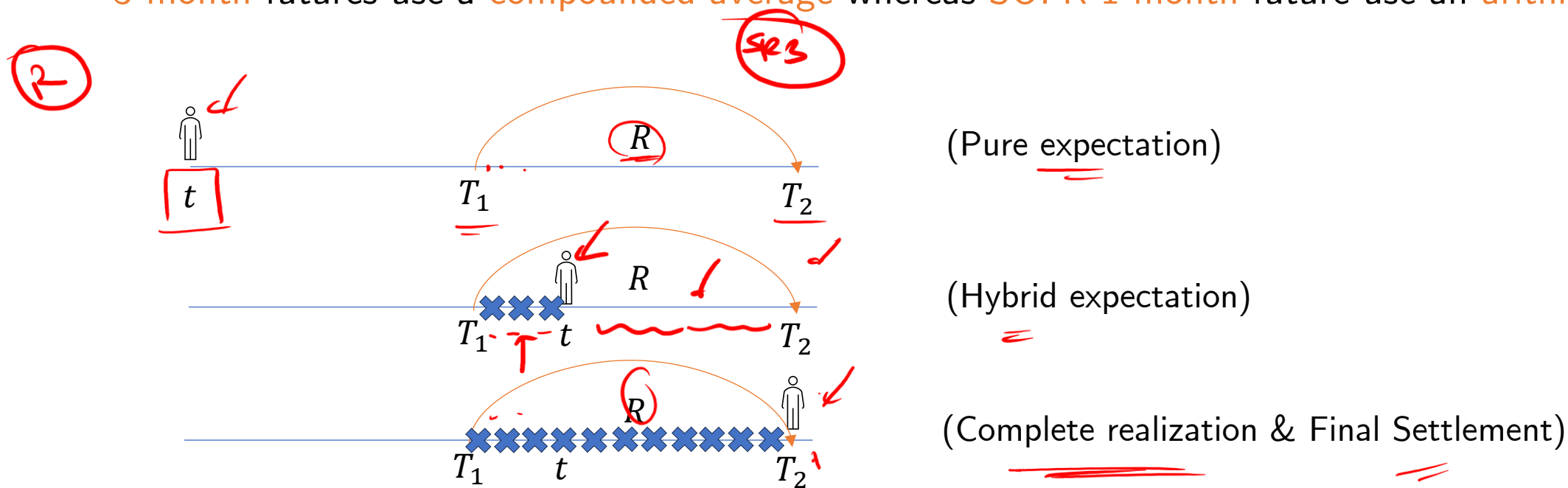
Number of Futures

10

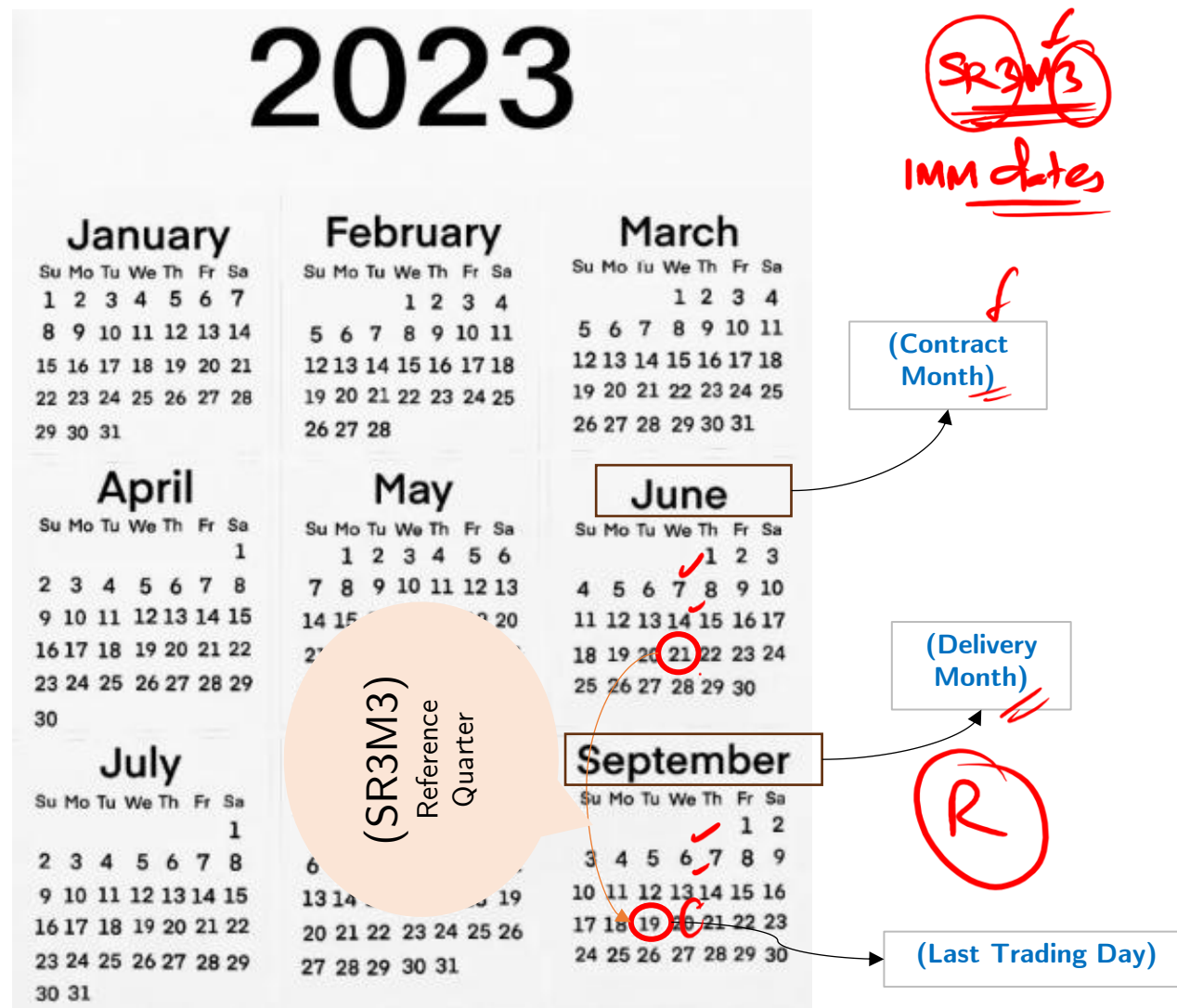
Date	Price	Rate (%)	+/- (bps)	Margin Balance	PnL
Mon, 20-Jun-22	99.00	1.00	1	\$ 25,200	\$ 25.00
Fri, 17-Jun-22	98.99	1.01	2	\$ 25,175	\$ 50.00
Thu, 16-Jun-22	98.97	1.03	-1	\$ 25,125	\$ (25.00)
Wed, 15-Jun-22	98.98	1.02	0	\$ 25,150	\$ -
Tue, 14-Jun-22	98.98	1.02	-5	\$ 25,150	\$ (125.00)
Mon, 13-Jun-22	99.03	0.97	-2	\$ 25,275	\$ (50.00)
Fri, 10-Jun-22	99.05	0.95	-1	\$ 25,325	\$ (25.00)
Thu, 9-Jun-22	99.06	0.94	3	\$ 25,350	\$ 75.00
Wed, 8-Jun-22	99.03	0.97	3	\$ 25,275	\$ 75.00
Tue, 7-Jun-22	99.00	1.00	2	\$ 25,200	\$ 50.00
Mon, 6-Jun-22	98.98	1.02	1	\$ 25,150	\$ 25.00
Fri, 3-Jun-22	98.97	1.03	-5	\$ 25,125	\$ (125.00)
Thu, 2-Jun-22	99.02	0.98	2	\$ 25,250	\$ 50.00
Wed, 1-Jun-22	99.00	1.00	8	\$ 25,200	\$ 200.00
Tue, 31-May-22	98.92	1.08	-	\$ 25,000	
Current Price	99.00				
Start Price	98.92				
Net Margin	\$ 200.00				
		Net Margin			
		\$ 200.00			

Rate implied by SOFR Future

- ▶ What does the reference rate underlying a SOFR Future contract represent ? It represents the market expectation of the term rate for the contract period. However, the realized term rate that is part of the final settlement won't be fully known until the end of the contract. So, as one enters the reference period, the term rate implied by SOFR future represents a hybrid of realized SOFR rates up to that point and market expectation of the rates for the remaining period.
- ▶ Note that the term rate for settlement is calculated using average daily rates over the period. **SOFR 3-month** futures use a **compounded average** whereas **SOFR 1-month** future use an **arithmetic average**.



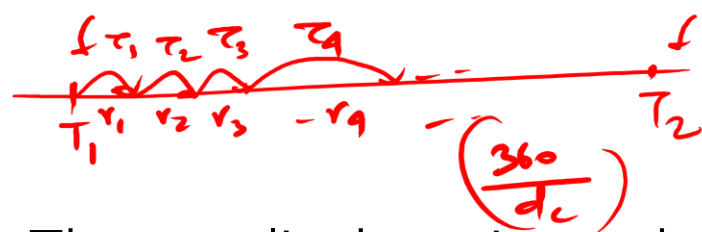
SOFR 3-month Future – Contract period



- ▶ **Reference Quarter** – The 3-month SOFR future follows the IMM calendar (3rd Wednesday of Mar, Jun, Sep, Dec). The Reference Quarter is the interval from the 3rd Wednesday (inclusive) of the month three months prior to the delivery month, to the 3rd Wednesday (exclusive) of the delivery month.
- ▶ For example, the product SR3M3 spans from 21st June 2023 (Inclusive) to 19th September 2023 (exclusive)

SOFR 3-month Future – Settlement

- ▶ The final settlement is based on compounded average of daily rates in the reference period. The handling of non-business days follows the standard SOFR compounding convention i.e.,

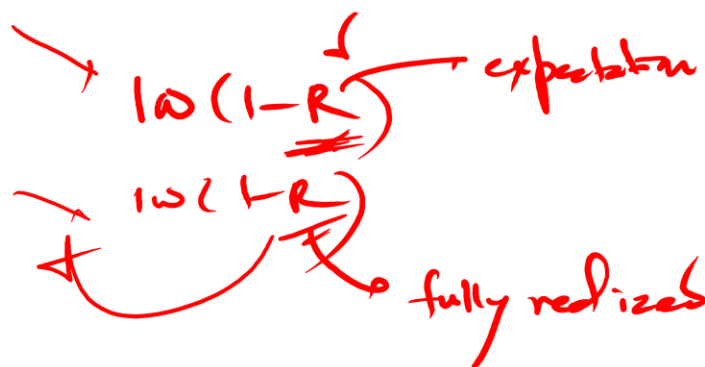


$$R = \left(\prod_{i=1}^{N_b} (1 + \underline{\underline{r_i \tau_i}}) - 1 \right) / \tau_c$$

100(1-R)

- ▶ The annualized rate is rounded to 1/100th basis point. The final settlement price is calculated as

$$\text{Price Basis} = 100(1 - R)$$



→ June 2025

SOFR 3-month Future – Settlement (Example)

Annualized Rate
Round to 1/100th BP
Settlement Price

5.239595%

5.23960%

94.7604

$$R = \left(\prod_{i=1}^{N_b} (1 + r_i \tau_i) - 1 \right) / \tau_c = \left(\frac{\text{Index}_t}{\text{Index}_{t_0}} - 1 \right) \times \left(\frac{360}{d_c} \right)$$

Value Date	Publication Date	Value Day	SOFR (%)	d_i	τ_i	$r_i \tau_i$	$(1 + r_i \tau_i)$
6/21/2023	6/22/2023	Wednesday	5.05%	1	0.002778	0.0001403	1.00014028
6/22/2023	6/23/2023	Thursday	5.05%	1	0.002778	0.0001403	1.00014028
6/23/2023	6/26/2023	Friday	5.05%	3	0.008333	0.0004208	1.00042083
6/26/2023	6/27/2023	Monday	5.05%	1	0.002778	0.0001403	1.00014028
6/27/2023	6/28/2023	Tuesday	5.05%	1	0.002778	0.0001403	1.00014028
6/28/2023	6/29/2023	Wednesday	5.06%	1	0.002778	0.0001406	1.00014056
6/29/2023	6/30/2023	Thursday	5.06%	1	0.002778	0.0001406	1.00014056
6/30/2023	7/3/2023	Friday	5.09%	3	0.008333	0.0004242	1.00042417
7/3/2023	7/5/2023	Monday	5.06%	2	0.005556	0.0002811	1.00028111
7/5/2023	7/6/2023	Wednesday	5.06%	1	0.002778	0.0001406	1.00014056
7/6/2023	7/7/2023	Thursday	5.06%	1	0.002778	0.0001406	1.00014056
7/7/2023	7/10/2023	Friday	5.06%	3	0.008333	0.0004217	1.00042167
7/10/2023	7/11/2023	Monday	5.06%	1	0.002778	0.0001406	1.00014056
7/11/2023	7/12/2023	Tuesday	5.06%	1	0.002778	0.0001406	1.00014056
7/12/2023	7/13/2023	Wednesday	5.05%	1	0.002778	0.0001403	1.00014028
7/13/2023	7/14/2023	Thursday	5.06%	1	0.002778	0.0001406	1.00014056
7/14/2023	7/17/2023	Friday	5.05%	3	0.008333	0.0004208	1.00042083
7/17/2023	7/18/2023	Monday	5.06%	1	0.002778	0.0001406	1.00014056
7/18/2023	7/19/2023	Tuesday	5.06%	1	0.002778	0.0001406	1.00014056
7/19/2023	7/20/2023	Wednesday	5.05%	1	0.002778	0.0001403	1.00014028
7/20/2023	7/21/2023	Thursday	5.06%	1	0.002778	0.0001406	1.00014056
7/21/2023	7/24/2023	Friday	5.05%	3	0.008333	0.0004208	1.00042083
7/24/2023	7/25/2023	Monday	5.05%	1	0.002778	0.0001403	1.00014028
7/25/2023	7/26/2023	Tuesday	5.06%	1	0.002778	0.0001406	1.00014056
7/26/2023	7/27/2023	Wednesday	5.06%	1	0.002778	0.0001406	1.00014056

SOFR 1-month Future – Contract Period

- **Contract Period** – The 1-month SOFR future spans all the calendar days of a month which is called the delivery month.

2023

January

Su	Mo	Tu	We	Th	Fr	Sa
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

February

Su	Mo	Tu	We	Th	Fr	Sa
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

March

Su	Mo	Tu	We	Th	Fr	Sa
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

April

Su	Mo	Tu	We	Th	Fr	Sa
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

May

Su	Mo	Tu	We	Th	Fr	Sa
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

June

Su	Mo	Tu	We	Th	Fr	Sa
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

July

Su	Mo	Tu	We	Th	Fr	Sa
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

August

Su	Mo	Tu	We	Th	Fr	Sa
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

September

Su	Mo	Tu	We	Th	Fr	Sa
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

(Delivery Month)

SOFR 1-month Future – Settlement



- ▶ The final settlement is based on arithmetic average of daily rates in the reference period. For non-business days the prevailing SOFR rate applies.

$$R = \left(\sum_{i=1}^{N_b} r_i \tau_i \right) / \tau_c$$

- ▶ The annualized rate is rounded to 1/10th basis point. The final settlement price is calculated as

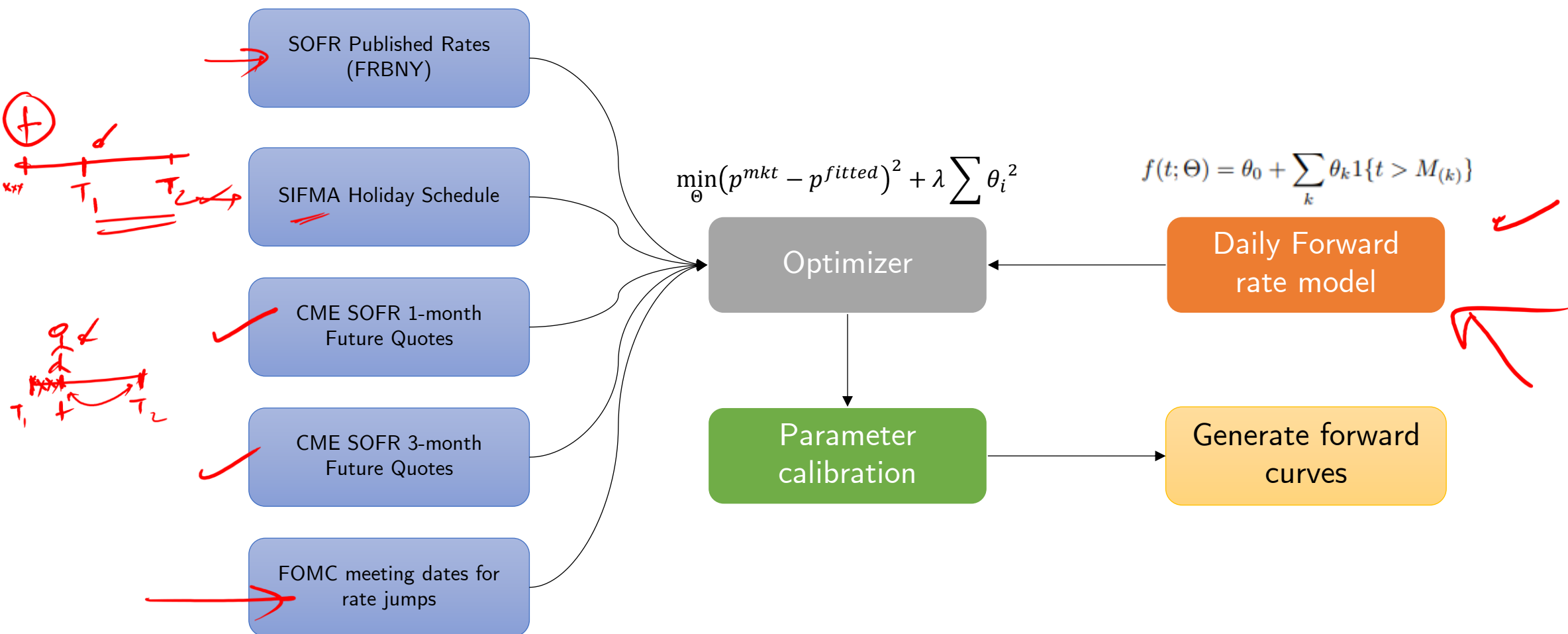
$$\text{Price Basis} = 100(1 - R)$$

SOFR 1-month Future – Settlement (Example)

Annualized Rate	5.098387%
Round to 1/10th BP	5.09800%
Settlement Price	94.902

Value Date	Publication Date	Value Day	SOFR (%)
7/1/2023	7/2/2023	Saturday	5.09%
7/2/2023	7/3/2023	Sunday	5.09%
7/3/2023	7/4/2023	Monday	5.06%
7/4/2023	7/5/2023	Tuesday	5.06%
7/5/2023	7/6/2023	Wednesday	5.06%
7/6/2023	7/7/2023	Thursday	5.06%
7/7/2023	7/8/2023	Friday	5.06%
7/8/2023	7/9/2023	Saturday	5.06%
7/9/2023	7/10/2023	Sunday	5.06%
7/10/2023	7/11/2023	Monday	5.06%
7/11/2023	7/12/2023	Tuesday	5.06%
7/12/2023	7/13/2023	Wednesday	5.05%
7/13/2023	7/14/2023	Thursday	5.06%
7/14/2023	7/15/2023	Friday	5.05%
7/15/2023	7/16/2023	Saturday	5.05%
7/16/2023	7/17/2023	Sunday	5.05%
7/17/2023	7/18/2023	Monday	5.06%
7/18/2023	7/19/2023	Tuesday	5.06%
7/19/2023	7/20/2023	Wednesday	5.05%
7/20/2023	7/21/2023	Thursday	5.06%
7/21/2023	7/22/2023	Friday	5.05%
7/22/2023	7/23/2023	Saturday	5.05%
7/23/2023	7/24/2023	Sunday	5.05%
7/24/2023	7/25/2023	Monday	5.05%
7/25/2023	7/26/2023	Tuesday	5.06%
7/26/2023	7/27/2023	Wednesday	5.06%

Overview of components



SOFR rates and data prep

	O	P	Q	R	S	T	U	V
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								
21								
22								
23								
24								
25								
26								
27								
28								
29								
30								
31								
32								

SOFR RATES
 (represents FOMC rate Jumps)

(Business Days SOFR rates)

Effective Date	SOFR	d	DIAP
12/2/2024	4.64%	1	1.000129
12/3/2024	4.64%	1	1.000129
12/4/2024	4.59%	1	1.000128
12/5/2024	4.59%	1	1.000128
12/6/2024	4.60%	3	1.000383
12/9/2024	4.63%	1	1.000129
12/10/2024	4.64%	1	1.000129
12/11/2024	4.62%	1	1.000128
12/12/2024	4.62%	1	1.000128
12/13/2024	4.60%	3	1.000383
12/16/2024	4.65%	1	1.000129
12/17/2024	4.62%	1	1.000128
12/18/2024	4.57%	1	1.000127
12/19/2024	4.30%	1	1.000119
12/20/2024	4.30%	3	1.000358
12/23/2024	4.31%	1	1.000120
12/24/2024	4.40%	2	1.000244
12/26/2024	4.53%	1	1.000126
12/27/2024	4.46%	3	1.000372
12/30/2024	4.37%	1	1.000121
12/31/2024	4.49%	2	1.000249
1/2/2025	4.40%	1	1.000122
1/3/2025	4.31%	3	1.000359
1/6/2025	4.27%	1	1.000110

(All days SOFR rates)

Effective Date	SOFR
3/1/2025	4.39%
3/2/2025	4.39%
3/3/2025	4.33%
3/4/2025	4.33%
3/5/2025	4.34%
3/6/2025	4.35%
3/7/2025	4.34%
3/8/2025	4.34%
3/9/2025	4.34%
3/10/2025	4.29%
3/11/2025	4.29%
3/12/2025	4.29%
3/13/2025	4.29%
3/14/2025	4.29%
3/15/2025	4.29%
3/16/2025	4.29%
3/17/2025	4.29%
3/18/2025	4.29%
3/19/2025	4.29%
3/20/2025	4.32%
3/21/2025	4.32%
3/22/2025	4.32%
3/23/2025	4.32%
3/24/2025	4.32%

SIFMA HOLIDAYS

1/1/2025
1/20/2025
2/17/2025
4/18/2025
5/26/2025
6/19/2025
7/4/2025
9/1/2025
10/13/2025
11/11/2025
11/27/2025
12/25/2025
1/1/2026
1/19/2026
2/16/2026
5/25/2026
6/19/2026
7/3/2026
9/7/2026
10/12/2026
11/11/2026
11/26/2026
12/25/2026
1/1/2027

- ▶ SOFR 3M Future uses compounded average of SOFR rates between IMM dates whereas SOFR 1M Future uses arithmetic mean of the SOFR rates for the delivery month
- ▶ Separate data preparation for SOFR 3M Future valuation vs SOFR 1M Future valuation considering different handling of non-business days (use SIFMA holiday schedule)
- ▶ Highlight FOMC rate jump dates

<https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm>

<https://www.sifma.org/resources/general/holiday-schedule/>

CME SOFR Future price

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 MAR 2025 SR1H5	 OPT		95.6725	UNCH (UNCH)	95.6725	95.675	95.675	95.67	12,188	09:05:26 CT 11 Mar 2025
 APR 2025 SR1J5	 OPT		95.69	UNCH (UNCH)	95.69	95.695	95.695	95.685	35,230	09:05:26 CT 11 Mar 2025
 MAY 2025 SR1K5	 OPT		95.78	UNCH (UNCH)	95.78	95.795	95.81	95.76	40,907	09:05:54 CT 11 Mar 2025
 JUN 2025 SR1M5	 OPT		95.875	-0.005 (-0.01%)	95.88	95.89	95.925	95.855	8,306	09:05:58 CT 11 Mar 2025
 JUL 2025 SR1N5	 OPT		96.00	UNCH (UNCH)	96.00	96.015	96.055	95.97	13,460	09:05:54 CT 11 Mar 2025

THREE-MONTH SOFR FUTURES - QUOTES

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MONTH	OPTIONS	CHART	LAST	CHANGE	PRIOR SETTLE	OPEN	HIGH	LOW	VOLUME	UPDATED
 DEC 2024 SR3Z4	 OPT		95.63	-0.0025 (UNCH)	95.6325	95.6325	95.6325	95.63	40,074	09:01:52 CT 11 Mar 2025
 JAN 2025 SR3F5	 OPT		-	-	95.65	-	-	-	0	08:48:31 CT 11 Mar 2025
 FEB 2025 SR3G5	 OPT		-	-	95.675	-	-	-	0	08:48:31 CT 11 Mar 2025
 MAR 2025 SR3H5	 OPT		95.7175	-0.0025 (UNCH)	95.72	95.7225	95.7375	95.705	244,957	09:05:57 CT 11 Mar 2025
 APR 2025 SR3J5	 OPT		-	-	95.81	-	-	-	53	08:48:31 CT 11 Mar 2025
 MAY 2025 SR3K5	 OPT		95.915	-0.015 (-0.02%)	95.93	95.915	95.915	95.915	27	09:00:30 CT 11 Mar 2025

SOFR FUTURE PRICES

Today

3/10/2025

1M SOFR Futures

#	MONTH	YEAR	PRODUCT	pMarket
1	March	2025	SR1H5	95.67
2	April	2025	SR1J5	95.69
3	May	2025	SR1K5	95.78
4	June	2025	SR1M5	95.87
5	July	2025	SR1N5	95.99
6	August	2025	SR1Q5	96.085
7	September	2025	SR1U5	96.145
8	October	2025	SR1V5	96.245
9	November	2025	SR1X5	96.33
10	December	2025	SR1Z5	96.385
11	January	2026	SR1F6	96.425

3M SOFR Futures

#	MONTH	YEAR	PRODUCT	pMarket
1	December	2024	SR3Z4	95.6325
2	March	2025	SR3H5	95.7075
3	June	2025	SR3M5	95.99
4	September	2025	SR3U5	96.25
5	December	2025	SR3Z5	96.4

► Obtain CME 1-month and 3-month SOFR Future quotes

Model Price for SOFR Futures

- ▶ Specify a model for forward dynamics (incorporating jumps on FOMC meeting dates)
- ▶ Price the CME SOFR Futures under specific averaging and non-business day handling needs for 1M and 3M Futures
- ▶ The rates used in the calculation will be a hybrid of realized SOFR rates and forward rates from the model

FOMC Jump dates	θ
	4.290% → θ_0
3/20/2025	0.028%
5/8/2025	-0.122%
6/20/2025	-0.175%
7/31/2025	-0.097%
9/18/2025	-0.155%
10/30/2025	-0.090%
12/11/2025	-0.097%

Daily Forward Rate Model

$$f(t; \Theta) = \theta_0 + \sum_k \theta_k 1\{t > M_{(k)}\}$$

Optimization

	A	B	C	D	E	F	G	H
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SOFR FUTURE PRICES
 Today

1M SOFR Futures

#	MONTH	YEAR	PRODUCT	p^{Market}	p^{fitted}	Sq. error
1	March	2025	SR1H5	95.67	95.68	0.00014
2	April	2025	SR1J5	95.69	95.68	0.00006
3	May	2025	SR1K5	95.78	95.78	0.00001
4	June	2025	SR1M5	95.87	95.87	0.00000
5	July	2025	SR1N5	95.99	95.98	0.00006
6	August	2025	SR1Q5	96.085	96.08	0.00007
7	September	2025	SR1U5	96.145	96.14	0.00000
8	October	2025	SR1V5	96.245	96.24	0.00006
9	November	2025	SR1X5	96.33	96.32	0.00007
10	December	2025	SR1Z5	96.385	96.39	0.00001
11	January	2026	SR1F6	96.425	96.42	0.00004

3M SOFR Futures

#	MONTH	YEAR	PRODUCT	p^{Market}	p^{fitted}	Sq. error
1	December	2024	SR3Z4	95.6325	95.6362	0.00001
2	March	2025	SR3H5	95.7075	95.7149	0.00005
3	June	2025	SR3M5	95.99	96.0072	0.00030
4	September	2025	SR3U5	96.25	96.2667	0.00028
5	December	2025	SR3Z5	96.4	96.4031	0.00001

- ▶ Minimize the squared error between Market quotes and Model based Prices
- ▶ Include a regularization term to penalize high values of jumps (even out jumps on FOMC dates)

Objective

$$\min_{\theta} (p^{mkt} - p^{fitted})^2 + \lambda \sum \theta_i^2$$

$$\lambda = \text{10}$$

$$\text{Obj} = \text{0.019667}$$

Generate Forward Curves

- With the risk-neutral calibration of the forward rates, we can generate 1M, 3M and 6M SOFR forward rates

